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## **From Reactive to Proactive: Machine Learning-Powered Drilling Analytics for Operational Excellence**

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### **Abstract**

Drilling operations face substantial challenges from non-productive time (NPT) and wellbore instability, costing the industry over \$7 billion annually. Current optimization efforts are constrained by three key data challenges: (1) analytical complexity in processing historical data, (2) statistical limitations in pattern recognition, and (3) restricted access to relevant datasets. This study presents an integrated machine learning (ML) approach that transforms historical drilling data analysis from reactive troubleshooting to predictive risk management, enabling preemptive mitigation of instability events through data-driven decision making.

Traditional mitigation strategies are hampered by fragmented data systems, with nearly 40% of drilling data remaining underutilized (SPE, 2022), and reliance on reactive problem-solving. This study showcases how historical data sheds light on challenging formations, enabling proactive mitigation planning through exploring the statistical analysis of geomechanics studies, offset wells, and drilling parameters. The proposed methodology workflow adopts two-phase methodology to harness historical data for drilling optimization. First, a structured data integration process focusing on geomechanical parameters (e.g., WOB, RPM, mud weight, ROP, ECD, shale reactivity), stuck pipe incidents, and NPT logs; and Second, ML-driven predictive modeling employs ensemble machine learning methods (Gradient Boosting, Random Forest) versus linear regression trained on offset data to forecast instability risks.

The study outputs emphasize the critical role of structured data management in handling overwhelming historical datasets, focusing on specific drilling events to inform future well designs. Analyzing multiple sets of offset wells before new drilling projects, the study unveils strategies to address issues like stuck pipe due to wellbore instability, clay swelling, and NPT reduction from drilling operations in the Middle East and Asia. This study illuminates the optimization of diverse drilling scenarios, including gas wells, clay formations, and laminar shaly formations through the application of historical statistical geomechanics studies, showcasing the transformative role of machine learning and data analytics in enhancing drilling performance.

Key outcomes include: (1) validated strategies for mitigating stuck pipe incidents and clay swelling, (2) a real-time recommendation engine for dynamic parameter adjustment (WOB, RPM, mud weight), and (3) significant NPT reduction through structured data management. Field applications of this framework

across high-risk wells demonstrated significant improvements in drilling efficiency and cost reduction. The synergy between geomechanics studies and historical data analysis, empowered by machine learning, emerges as a pivotal component. This holistic partnership forms a beacon navigating the complexities of drilling optimization, paving the way for operational excellence.

This study provides a roadmap for industry applications, showcasing the symbiotic relationship between geomechanics studies and ML-driven historical data analysis to enhance operational excellence and minimize NPT in drilling projects.

**Key Words:** Drilling Analytics, Machine Learning, Geomechanics, NPT Reduction, Predictive Modeling, Automated Drilling Optimization

## Introduction

The development of wells in geologically complex formations increasingly relies on the systematic analysis of historical drilling data (Omer et al. 2023). A foundational step in this process is to develop a thorough understanding of the available datasets and structure them in a way that enables meaningful interpretation and supports informed well planning. Gala et al. (2010) identified prevalent drilling challenges, and the mitigation strategies commonly adopted, highlighting the critical role of engineering evaluation in improving drilling efficiency. In a related study, AlBar et al. (2018) emphasized that constructing a real-time data analytics environment for drilling is inherently multifaceted, requiring a robust data engineering pipeline capable of integrating diverse datasets across multiple wells and fields. This integration facilitates accelerated analysis and continuous knowledge extraction from historical operations. Planning future drilling centers involves navigating a highly complex decision space with numerous potential configurations, where manual screening and optimization become impractical due to the vast data volume and multifaceted operational constraints (Salimov et al. 2025). Advanced AI techniques—such as machine learning-driven optimization, constraint-aware algorithms, and intelligent decision-support systems—are essential to efficiently explore solution spaces and identify optimal strategies.

More recently, the integration of artificial intelligence (AI) and machine learning techniques has further enhanced this analytical framework. These technologies enable predictive modeling, automated pattern recognition, and data-driven insights that support optimization of drilling operations (Gamal et al. 2023).

As shown in Figure 1, the workflow for advanced drilling analytics begins with the systematic identification of relevant offset wells, followed by the acquisition of their associated datasets. These datasets undergo stringent data quality audits designed to detect and eliminate incomplete, inconsistent, or unreliable records, thereby ensuring a robust and reliable foundation for subsequent analysis. Once validated, the refined data are integrated with real-time drilling measurements and geological information, forming a comprehensive dataset for analysis. This integrated dataset is processed using advanced artificial intelligence (AI) and machine learning (ML) frameworks, which employ supervised learning techniques to model historical drilling outcomes and identify patterns linked to operational performance and drilling hazards. Through pattern recognition algorithms, the system detects early signals of potential complications such as wellbore instability, lost circulation, tight hole conditions, and kick occurrences.

Ma et al. (2018) developed a methodology leveraging natural language processing (NLP) and deep neural networks (DNNs) to extract critical information from unstructured daily drilling reports (DDRs) from over 2000 wells. The approach integrated drilling and completion data to generate key operational and reservoir management metrics, the method reduced data processing time from months to days and enabled rapid identification of major NPT categories.

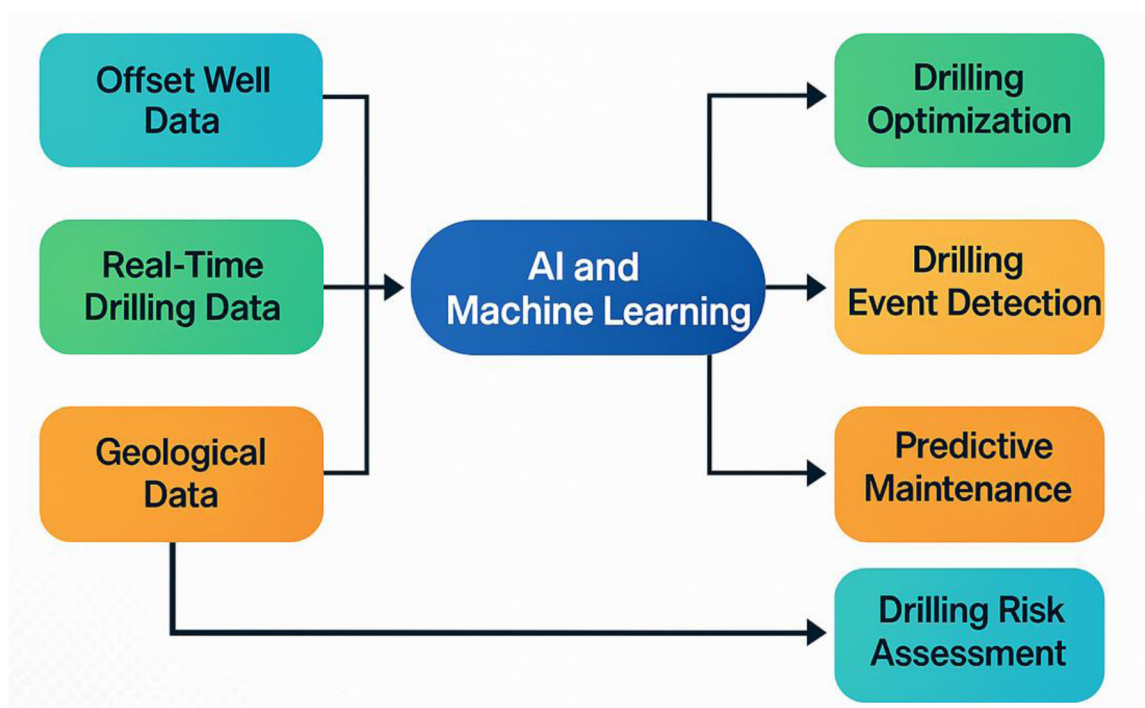


Figure 1—AI powered Workflow flow for Drilling Analytics

Curina et al. (2022) demonstrated the application of machine learning integrated with IoT-enabled data acquisition to identify rig inefficiencies and optimize operations. In their case study, rig data was continuously collected, standardized, and cleansed to remove noise, redundancy, and outliers, with feature selection applied to isolate key parameters critical for forecasting and optimization. These refined datasets were then processed through an in-house developed system that monitored equipment performance, particularly focusing on the Hydraulic Power Unit (HPU) powering a hydraulic hoisting rig, which exhibited low efficiency. Machine learning models were trained and tested on this curated data to evaluate the impact of varying operational parameters, with special attention to tripping speed and minor procedural adjustments that could enhance efficiency and minimize invisible lost time. The implementation of these insights enabled the rig to operate more efficiently, extend equipment lifespan, reduce downtime, and lower the risk of equipment failure while maintaining loads within design specifications.

### Key Applications & Features of Machine Learning in Drilling Oil and Gas Wells

- Real-time web-based visualization supported by dynamic models and continuously updated advanced algorithms.
- Physics-based engineering calculations and real-time analytics enable proactive, on-the-fly responses to emerging operational anomalies.
- Automated multi-source data integration facilitates optimized decision-making across all operational stages.
- Enhances individual engineering capabilities to match the performance of a multidisciplinary team of specialists.
- Physics-driven, automated calculations and adaptive workflows deliver continuous performance optimization.
- Enables second-by-second parameter adjustments to maximize downhole efficiency and productivity.
- Machine learning-driven models forecast potential risk exposures in real time.

- Supports reduction of non-productive time (NPT) and invisible lost time (ILT) through proactive mitigation strategies.
- AI-powered predictive engines ensure operations remain within technical limits while sustaining peak rate of penetration (ROP).
- Maintains well integrity and operational safety without compromising efficiency.
- Configurable thresholds generate automated notifications tailored to operational contexts.
- Provides early warning systems to enable rapid evasive actions and data-driven decision-making.

Additionally, predictive algorithms are applied to estimate key performance indicators, including rate of penetration (ROP) and downhole vibration detection during drilling operations (Al-Abduljabbar et al. 2020; Saadeldin et al. 2023). The implementation of an automated drilling process platform with an AI-powered ROP optimizer was carried out by Zhao et al. (2025) for ADNOC Offshore to fully automate the drill-a-stand process in an offshore UAE field which helped in significant on-bottom ROP improvements of 27%. Optimization of WOB and RPM parameters was achieved within the defined search space, replacing manual drill-off tests traditionally dependent on driller expertise—and conduct dynamic risk assessments in real time. By combining supervised learning with predictive modeling, AI-driven analytics enable a proactive approach to drilling operations, reducing non-productive time, enhancing safety, and optimizing overall efficiency (Gamal et al. 2024).

## Machine Learning Powered Drilling Analytics

The integration of Artificial Intelligence (AI) into drilling data analysis has substantially improved the efficiency and depth of subsurface evaluations (Gamal 2025). In this study, AI was utilized to process and interpret multidimensional datasets comprising formation-specific mud weight windows, associated drilling risks, and spatial well coordinates.

AI-powered tools enable rapid transformation of raw tabular data into insightful visualizations (Figure 2), such as mud weight envelope plots, risk distribution charts, and geospatial well proximity maps. These representations facilitated a clearer understanding of formation behavior and allowed for the identification of high-risk zones—such as intervals prone to losses, breakout events, or stuck pipe incidents.

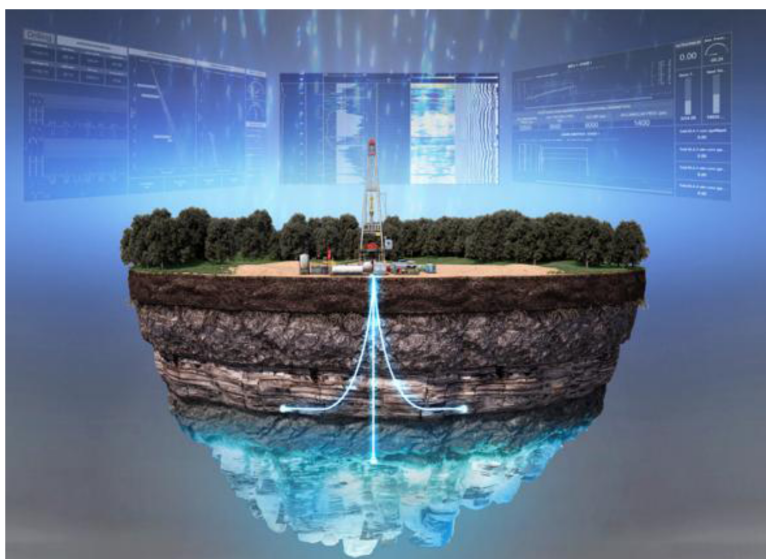


Figure 2—Re-imagining Drilling optimization using AI



Moreover, the classification and frequency mapping of recommended operational responses—like LCM deployment, ROP control, or ECD adjustments—were efficiently automated using AI-driven text parsing and pattern recognition. This significantly reduced the manual effort required for operational planning and allowed for a more structured approach to risk mitigation.

### Harnessing historical data for drilling optimization

This paper describes the drilling analytics and physics-based analyses in the exploratory vertical well in Asia. It was conducted to evaluate and prevent the risk of wellbore instability regarding drilling this well. Leveraging prior drilling experiences from offset wells is essential for diagnosing formation-specific challenges and understanding their interplay with controllable drilling parameters. Historical well data serves as a critical input to validate and calibrate geomechanical models prior to their application in future well planning, particularly for mud weight window optimization.

In this context, the analysis focused on extracting relevant operational insights from daily drilling reports (DDRs) of four nearby wells. These reports were reviewed in detail to identify occurrences of geomechanical-related issues such as stuck pipe, tight hole conditions, reaming, wellbore influx, and total losses. Alongside, actual mud weight values used during the drilling of each interval were recorded. The extracted data was compiled into mud weight versus measured depth plots to visually correlate operational problems with wellbore stability margins.

The use of spatial visualization, supported by AI, plays a vital role in offset well analysis and planning. By mapping the relative positions of nearby wells and overlaying historical drilling challenges, engineers can identify geospatial trends and potential risk zones around the planned well trajectory. AI enhances this process by analyzing large datasets to detect patterns in wellbore stability, pressure regimes, and operational hazards that may not be evident through traditional methods. This enables more informed decisions regarding well placement, mud weight design, and drilling practices, ultimately improving safety, reducing non-productive time (NPT), and optimizing overall well performance.

Figure 3 shows spatial visualization of offset wells that were selected. Well SH\_1 was closest to the target well with approximately 6km distance and DH\_1 was farthest from the target.

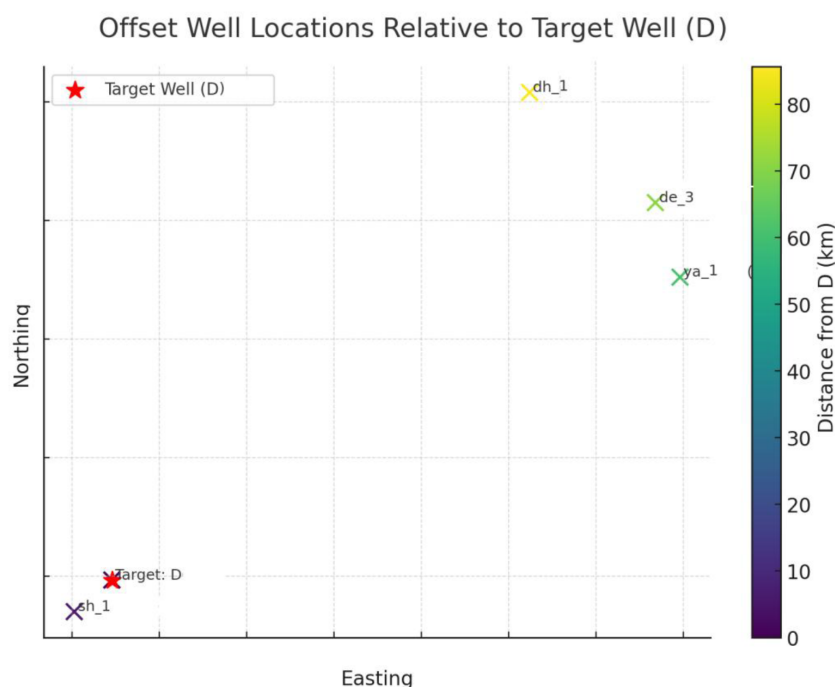


Figure 3—Spatial Mapping of Wells

Offset wells exhibiting comparable stratigraphic and structural settings will be carefully selected to ensure relevance in the geomechanical assessment. Particular attention must be given to potential compartmentalization within the reservoir, as it can significantly impact formation behavior and data applicability. The selection process should incorporate multiple criteria—such as formation continuity, faulting, and pressure regimes—to accurately capture geological and structural variations across the target field, thereby improving the reliability of subsequent modeling and analysis.

Figure 4 is a visual representation of the Mud Weight (MW) range for each formation. Each vertical line shows the minimum to maximum MW (ppg) for a formation, helping to easily compare mud weight requirements across formations. Figure 5 shows the frequency of different drilling risks across all formations. "Losses" is the most frequently encountered risk, followed by "Break outs" and "Stuck pipe".

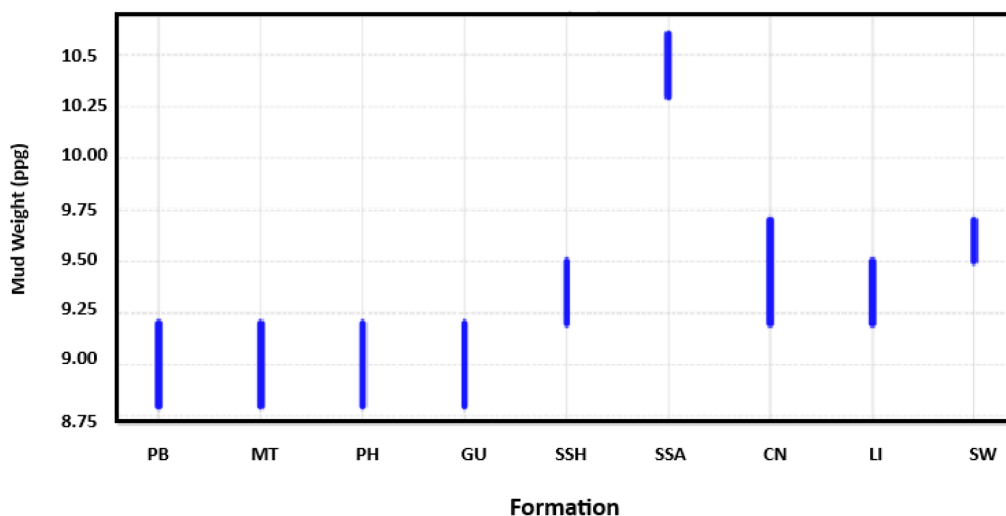


Figure 4—Mud Weight used to drill each formation

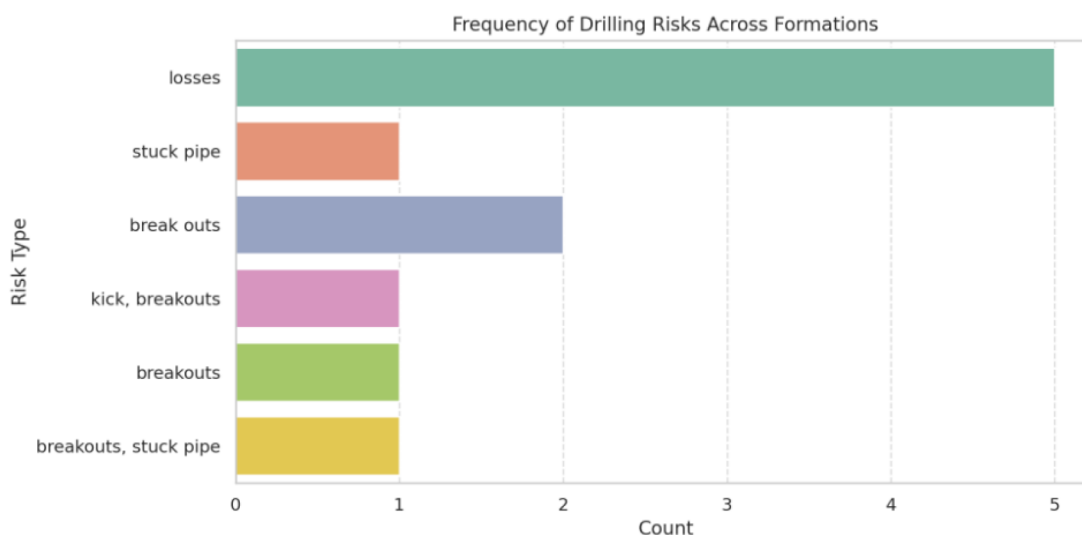


Figure 5—Drilling Events Distribution

The heatmap as shown in Figure 6 visually compares minimum and maximum Mud Weight (MW ppg) across formations. Darker colors indicate lower values, and lighter colors indicate higher values. The sandstone formation SSA required high mud weight to drill across the formation which stands out with the highest MW range (10.3–10.6 ppg) whereas PB, MT, PH, GU can be drilled with a mud weight range between 8.8 – 9.2 ppg.



Figure 6—Heatmap of Mud Weights

### Combining physics-driven models with AI to analyze rock mechanics in offset well formations

There are several un-certainties in the physics-based study including, assumption of 1-Dimensional isotropic and homogeneous model, no chemical rock fluid interaction is assumed; effect of discontinuities like bedding failure, highly laminated layers, etc., are not included.

The evaluation of horizontal and vertical stresses helps in planning the mud weight window to drill safe. From The radar chart [Figure 7](#) it is clear that formation GU has low maximum horizontal stress but high vertical stress.

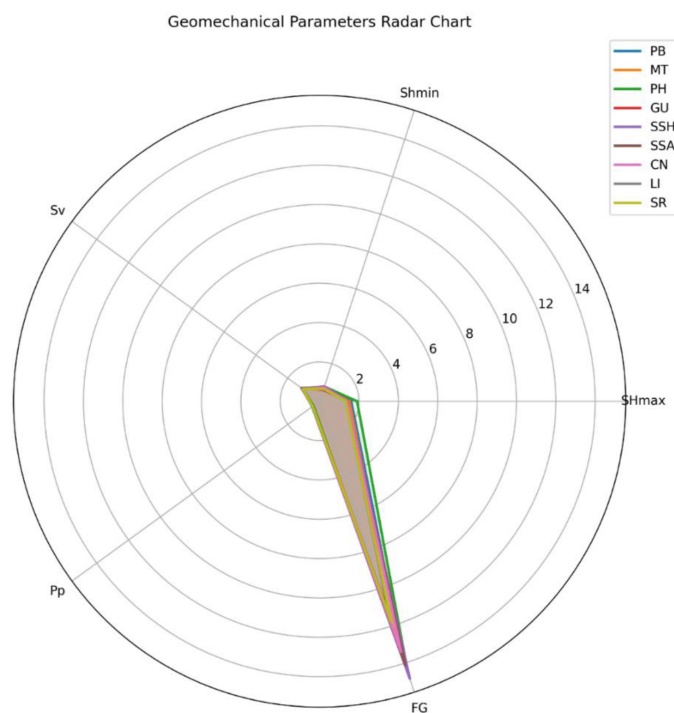


Figure 7—Radar Plot of Geomechanical Parameters

As shown in pore pressure versus FG plot ([Figure 8](#)), formation SSH has the highest FG (14.83 ppg) despite moderate pore pressure. This is Critical for mud weight selection to avoid fractures.

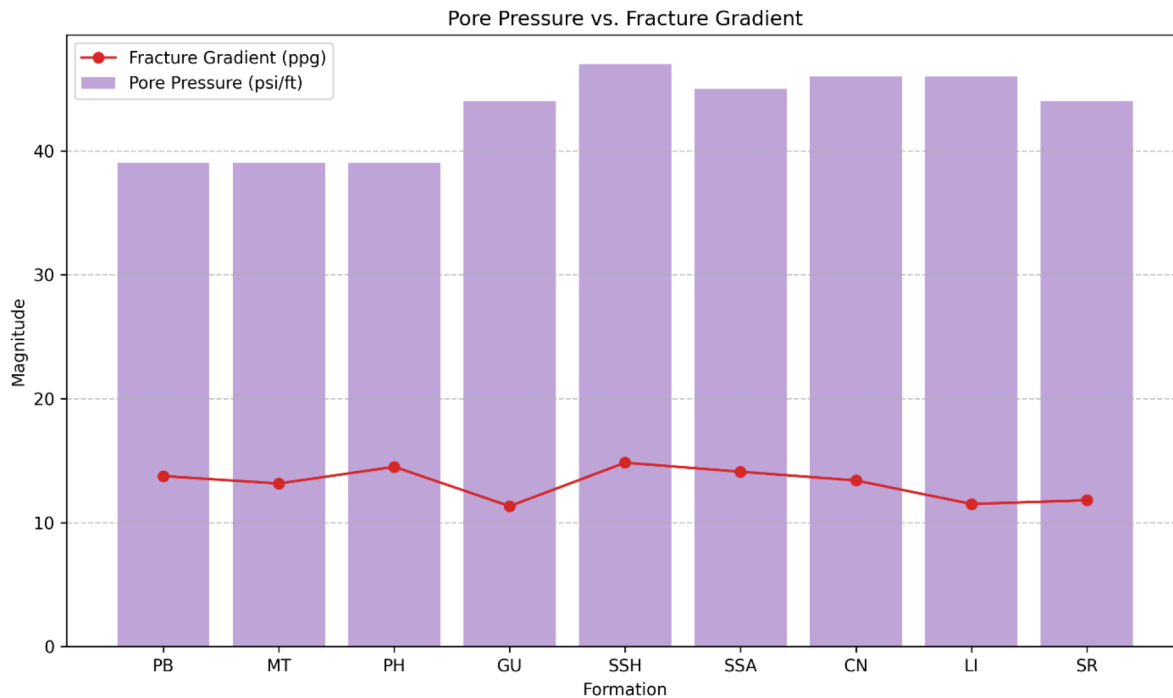


Figure 8—Pore Pressure Distribution across the formation

Formation **PH** has the highest SHmax i.e.1.9 psi/ft (Figure 9), indicating potential instability. This combination of hybrid physics based analysis with AI can be used to flag high-risk zones for drilling.

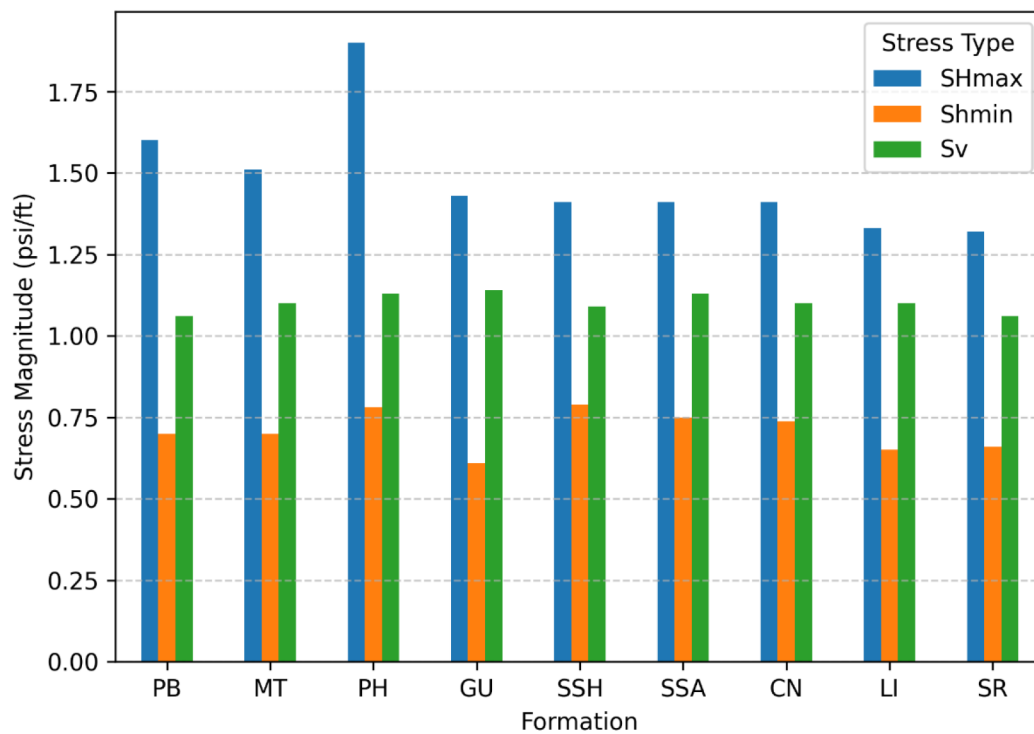


Figure 9—Principal Stress distribution

As shown in Figure 10 below formation PH shows high stress anisotropy ( $SH_{max}/Sh_{min} \approx 2.4$ ). This means that there is high risk of shear failure. We might want to design appropriate mud weight and use proper hole cleaning technical procedures



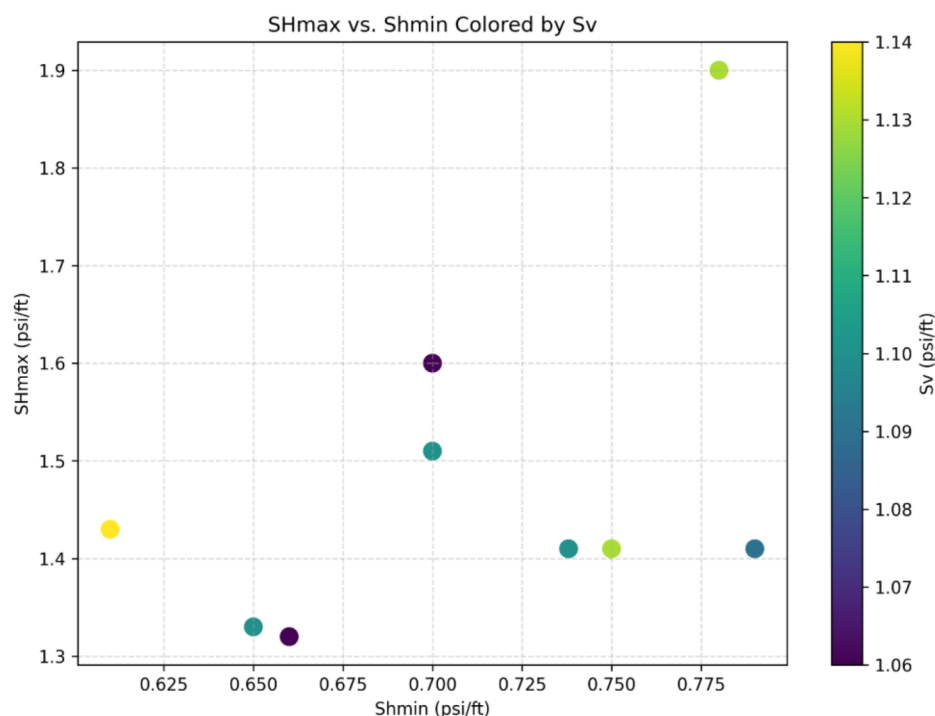


Figure 10—Stress anisotropy

We analyzed drilling events like stuck pipe, caving's, losses, etc. in the shaley formations (Figure 11) which were highly problematic. These formations were analyzed across four wells SH\_1, YA\_1, DH\_1, DE\_03. The AI was model was trained to focus on Mud weight (MW) adjustments and their correlation with drilling issues, Problem frequency by well and depth and Critical actions taken to mitigate risks.

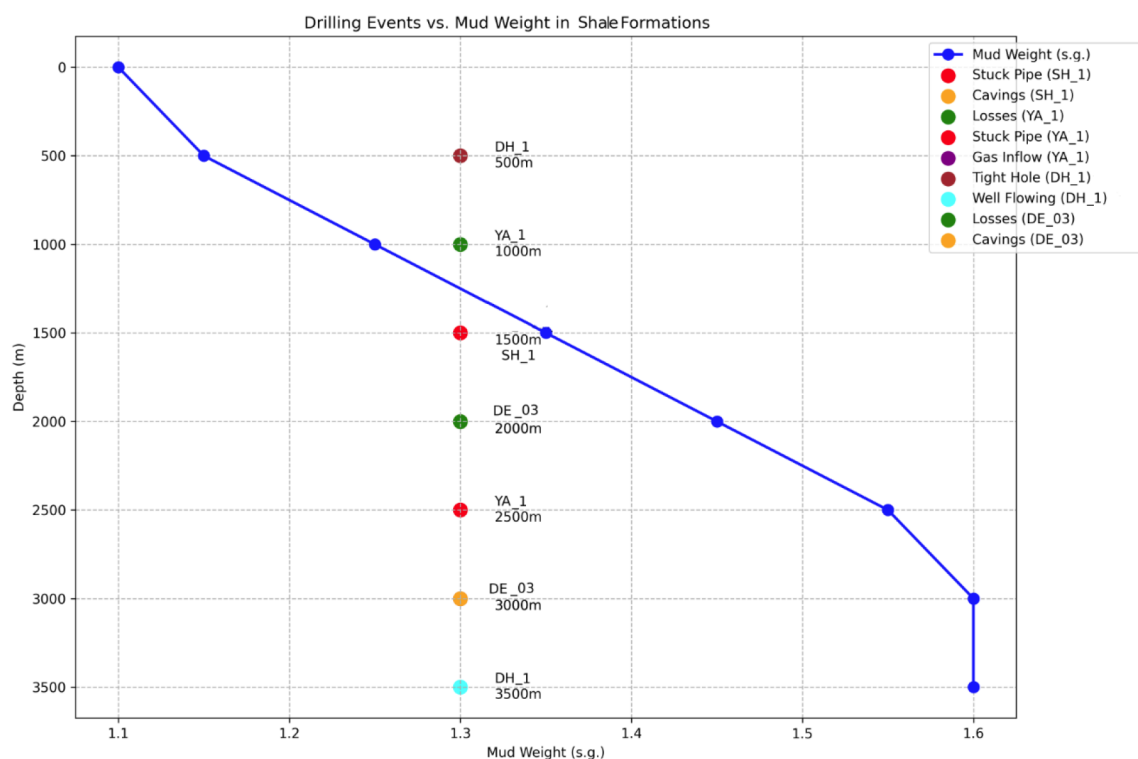


Figure 11—Drilling Challenges in Shaly formations

The AI based depth-based analysis helped in identifying Stuck pipe incidents which occurred at 1500m for well SH\_1 and 2500m for well YA\_1), prompting MW increases. Cavings were concentrated in shaly formation (2000–3000m) for almost all the wells, indicating wellbore instability. Losses at 1000m for well YA\_1) and 2000m for well DE\_03 required LCM (Lost Circulation Material) treatments.

A Problem Frequency Heatmap as shown in Figure 12 was generated by AI engine and it indicated that YA\_1 experienced the widest range of issues *losses, stuck pipe, gas inflow*), suggesting complex formation interactions. Well DH\_1 had recurring tight hole and cavings, highlighting mechanical instability.

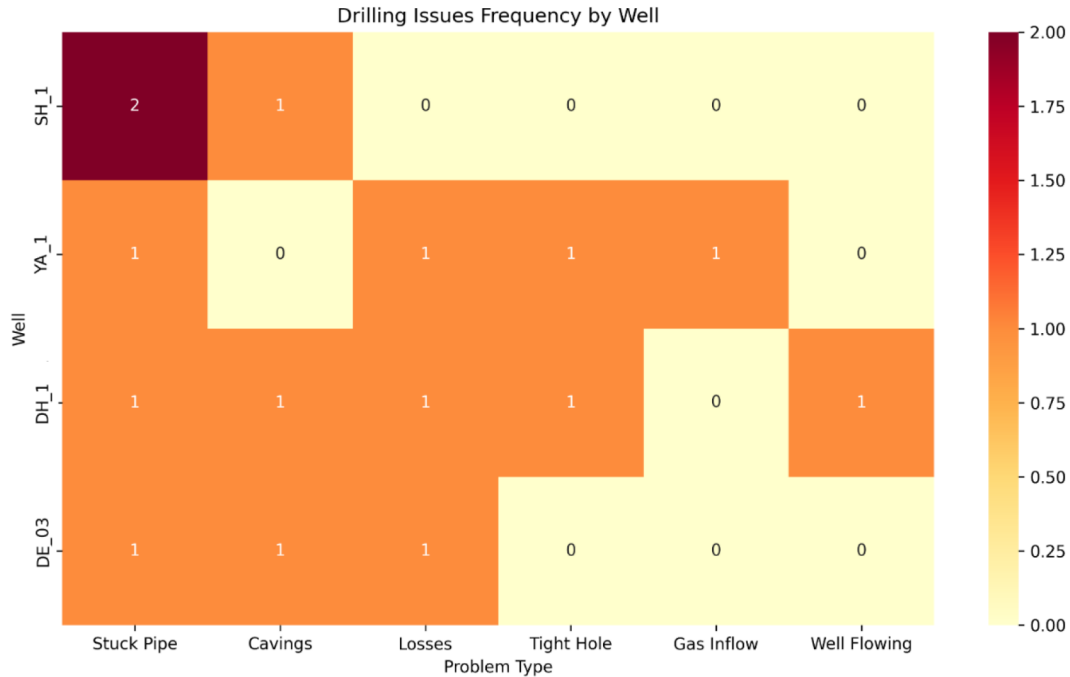


Figure 12—Frequency of Drilling Issue by Well

These observations underscore the value of integrating offset well experiences into the geomechanical assessment workflow, enabling more accurate mud weight design and improving drilling efficiency and safety in future operations.

Physics based 1D-Mechanical Earth Model for the well DG

Figure 13 presents the synthetic physics-based wellbore stability analysis for the proposed well DG, developed by integrating key rock mechanical parameters derived from offset wells SH\_1, YA\_1, and DH\_1. This integrated model includes estimates of critical geomechanical properties and in-situ stress field profiles spanning from the PB Formation down to the SW Formation, providing a comprehensive 1D Mechanical Earth Model (MEM) for the planned well trajectory.

The synthetic MEM captures formation-specific variations in rock strength, pore pressure, and stress gradients, thereby enabling a more reliable prediction of wellbore stability behavior (Omer et al. 2020). The model highlights that the SSH and SSA, LI, and SW formations are likely to pose well-bore stability challenges due to their comparatively weaker mechanical properties and higher tendency for instability under drilling conditions.

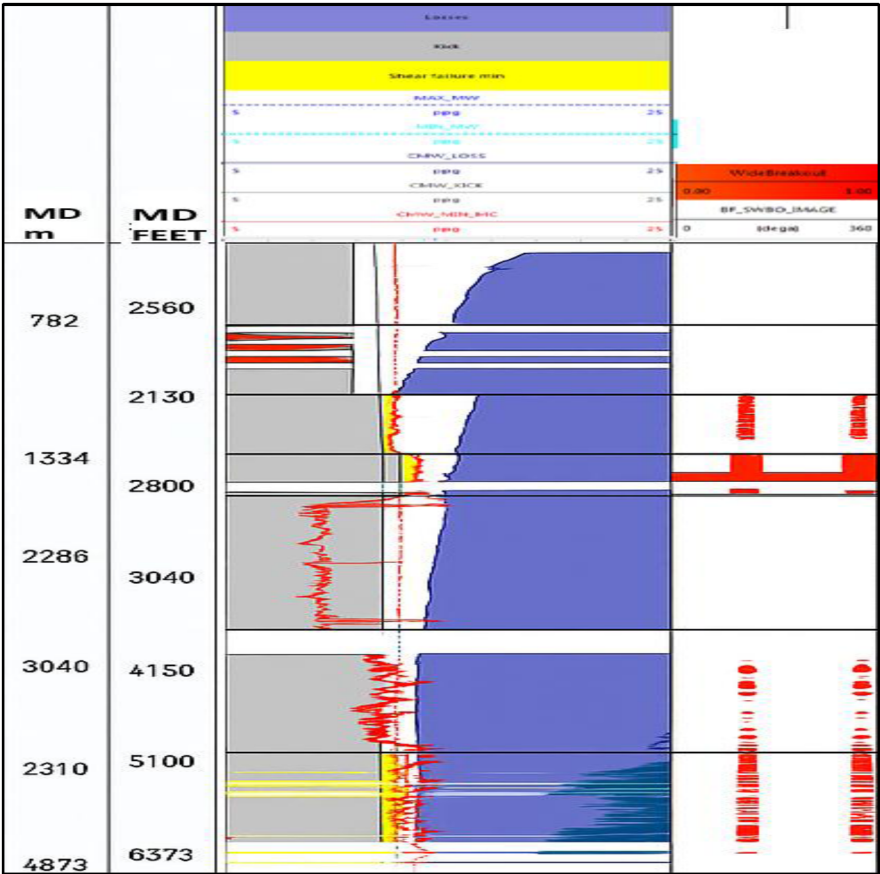


Figure 13—1D MEM for Target Well D

Machine Learning Modeling

Building the machine learning model to predict the safe mud weight from the input parameters was accomplished through comprehensive data cleaning, preprocessing, advanced data analytics for inputs versus output, training and testing the ML models and evaluating the ML key performance indicators (KPIs) in terms of prediction accuracy and errors between the actual versus predicted MW. The dataset includes parameters such as density (DEN), sonic data as compression and shear wave velocities (DTC/DTS), gamma ray (GR), pore pressure (Pp), maximum and minimum horizontal stresses (SHMAX, SHMIN), vertical stress (SVERTICAL), and unconfined compressive strength (UCS), alongside measured mud weight (MW) as the target variable.

A thorough exploratory data analysis was performed to understand the distributions and relationships between the measured parameters and the target variable (MW). Descriptive statistics, including mean, median, standard deviation, and range, were calculated for all input features. Scatter plots were generated for each input feature against MW to identify potential linear or nonlinear trends and to visually inspect the influence of different wells on data variability. A pairwise relationship analysis using seaborn's pairplot function was employed for the top five parameters to examine correlations and interactions between features, enabling an initial qualitative assessment of feature relevance as shown in Figure 14. The plots were created to visualize the selected features against MW for each well individually, as well as collectively for all wells. Color-coded visualizations facilitated differentiation among wells, highlighting inter-well variability and patterns within the dataset.

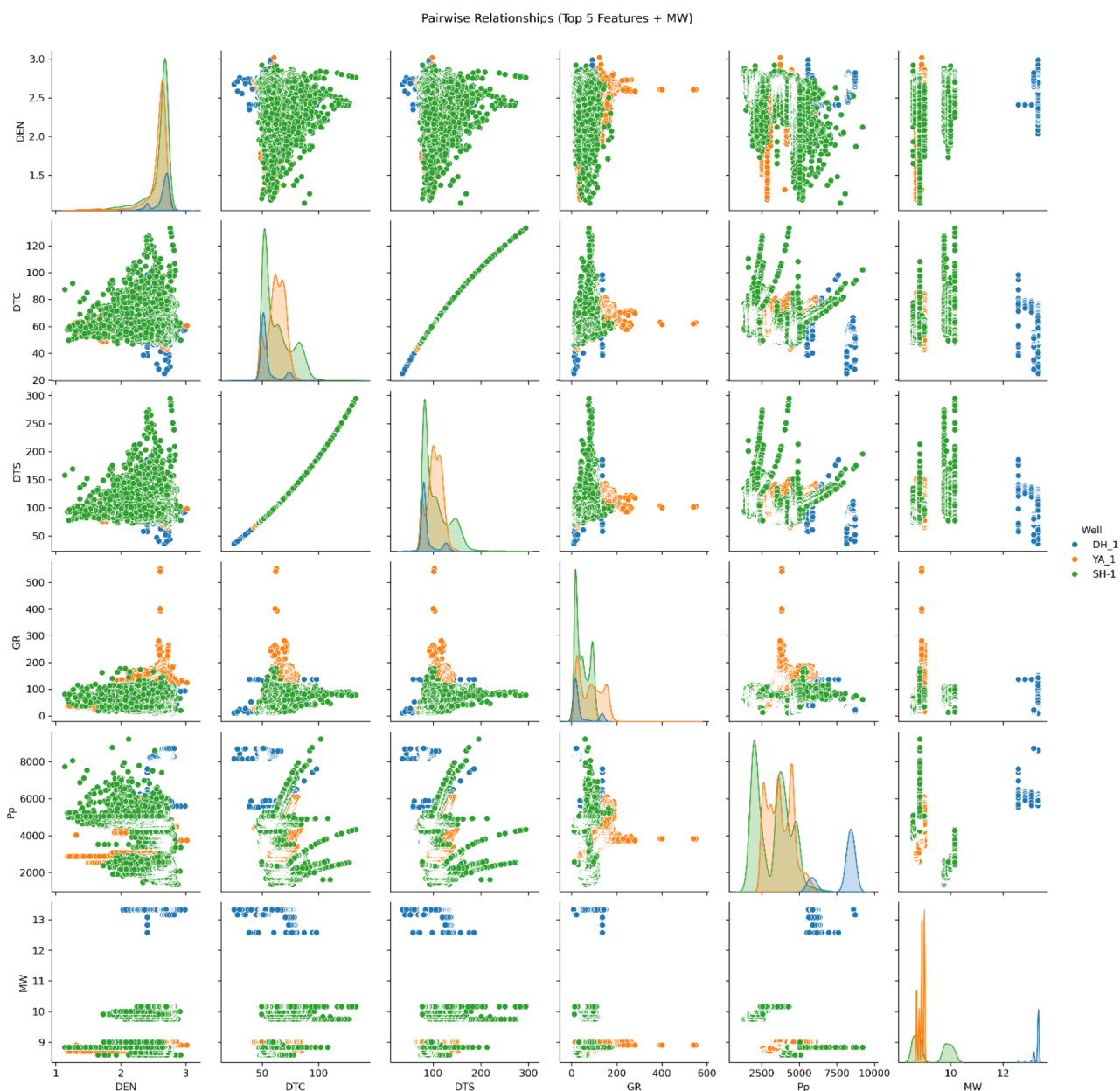


Figure 14—Data Visualizations

To reduce dimensionality and identify the most influential features affecting mud weight prediction, principal component analysis (PCA) was applied to the standardized input features. Standardization ensured all features contributed equally to the variance analysis, mitigating scale biases. PCA allowed the identification of a subset of principal features of the total variance, providing a reduced yet representative feature set for subsequent modeling. The contribution of each original feature to the principal components was evaluated to prioritize parameters with the highest predictive potential as represented in Figure 15. From all input parameters [pore pressure (Pp), gamma ray (GR), delta-t shear (DTS), delta-t compressional (DTC), density (DEN), minimum horizontal stress (SHMIN), maximum horizontal stress (SHMAX), uniaxial compressive strength (UCS), vertical stress (SVERTICAL)], only 5 parameters [‘Pp’, ‘GR’, ‘SVERTICAL’, ‘SHMIN’, ‘DTC’] were selected as main inputs to be fed to the ML models for training.

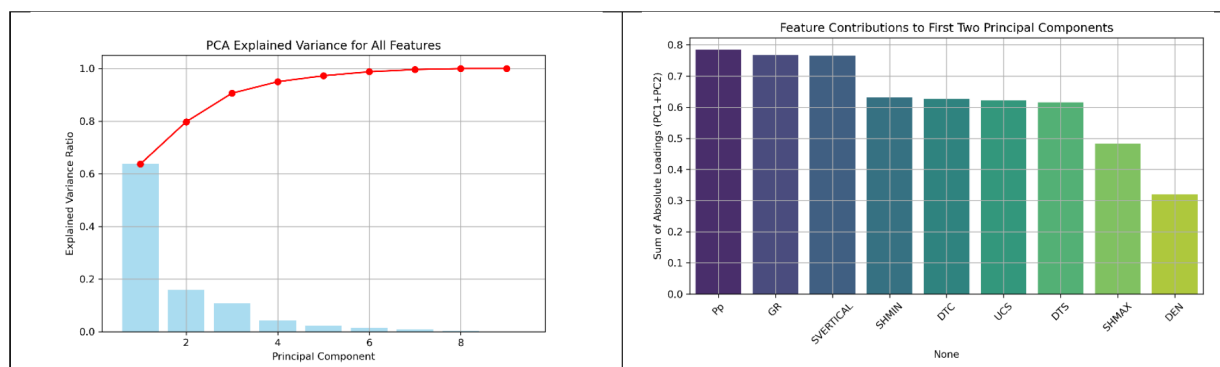


Figure 15—PCA

Following PCA, a correlation heatmap (Figure 16) was constructed for the selected features along with MW to quantitatively evaluate linear dependencies.

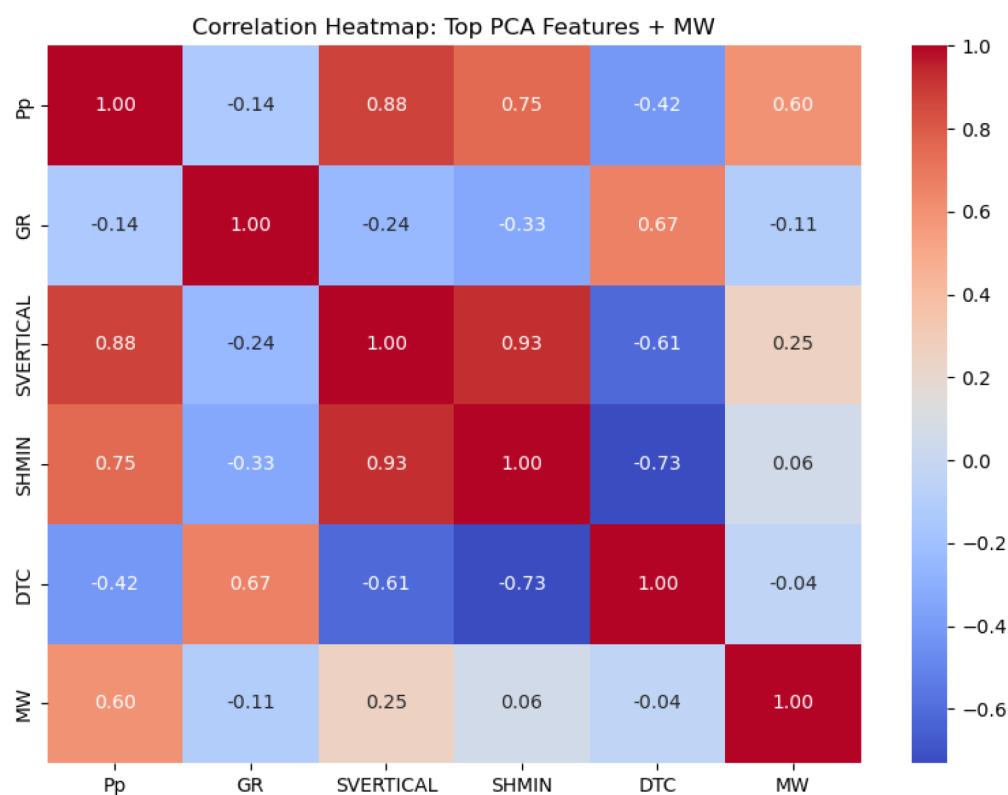


Figure 16—Correlation Heat Map

Machine learning (ML) techniques were employed to develop predictive models for mud weight estimation. Three algorithms were compared named Random Forest Regressor, Gradient Boosting Regressor, and Linear Regression. The workflow included:

For each well, the top PCA-selected features were used to train models on a 70:30 train-test split, ensuring adequate evaluation of model performance for intra-well prediction. Performance metrics such as  $R^2$  (coefficient of determination) and RMSE (root mean squared error) were computed, and predicted vs. actual mud weight plots were generated for visual validation. The ML results for the three algorithms are reported in Figure 17. Random Forest achieved the highest accuracy with an  $R^2$  of 0.9997 and a root mean squared error (RMSE) of 0.0225, demonstrating its exceptional ability to capture complex, nonlinear interactions among wellbore parameters. Gradient Boosting also performed very well, yielding  $R^2 = 0.9973$

and  $RMSE = 0.0681$ , slightly less precise than RF but still highly reliable. In contrast, Linear Regression produced moderate performance with  $R^2 = 0.7994$  and  $RMSE = 0.5855$ , reflecting its inability to fully model the nonlinear relationships inherent in drilling data. These results highlight the superiority of tree-based ensemble models for accurate MW prediction and emphasize the importance of feature selection through PCA to focus on the most informative parameters. The comparison underscores that while linear models provide interpretability, ensemble methods such as RF and GB offer robust predictive capability, crucial for real-time well control decision-making.

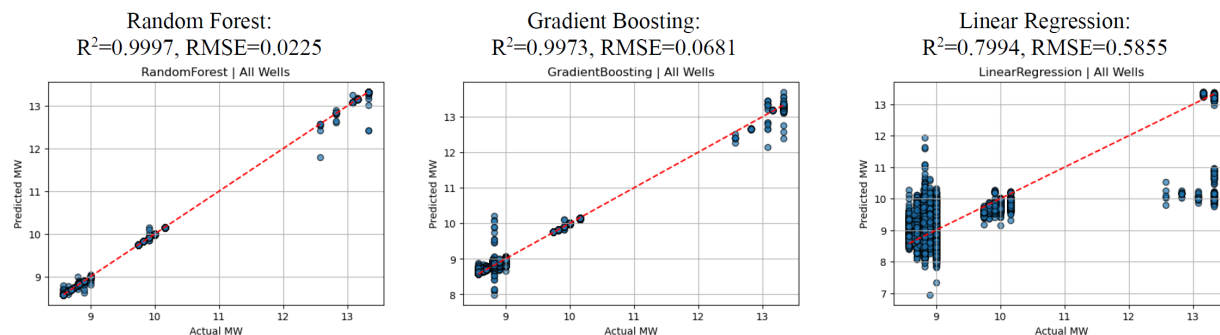


Figure 17—ML Results

## Conclusions and Future Recommendations

The integration of advanced digital technologies including machine learning (ML), data mining, and exploratory data analysis represents a pivotal advancement in the digital transformation of drilling and completion operations. These approaches enable rapid quality control, seamless data integration, and predictive analysis of large-scale datasets, accelerating the identification of operational challenges and supporting the optimization of drilling strategies. Beyond the demonstrated gains in performance, safety, and capital efficiency, the scalability and adaptability of these digital solutions position them as transformative tools with strong potential for widespread adoption across the industry, facilitating more resilient and data-driven reservoir and well management practices.

The ML methodology applied in this study combines thorough exploratory data analysis, dimensionality reduction via principal component analysis (PCA), advanced visualization, and supervised machine learning techniques to construct robust predictive models for mud weight estimation. This workflow systematically selects the most informative input parameters, provides insights into complex inter-parameter relationships, and evaluates both intra- and inter-well predictive capabilities using multiple ML models. Tree-based ensemble methods, particularly Random Forest and Gradient Boosting, were shown to outperform linear regression models, capturing nonlinear interactions among features and providing highly accurate mud weight predictions for individual wells and combined datasets.

The application of AI and ML proved instrumental in accelerating data-driven decision-making, allowing engineers to extract meaningful insights from complex drilling parameters in a fraction of the time required using traditional methods. This capability supports more proactive well planning, reduces non-productive time, and enhances overall operational safety and cost efficiency.

## Future Recommendations

- **Generalization to Larger Datasets:** Expanding the study to include a more extensive dataset with a wider range of wells and diverse geological conditions will improve the robustness and generalizability of the ML models.



- **Multi-Well and Real-Time Integration:** Developing adaptive, real-time predictive systems that continuously learn from new drilling data can enhance well control and integrity management across multiple wells simultaneously.
- **Enhanced Feature Engineering:** Incorporating additional real-time downhole measurements, drilling dynamics, and operational metadata could further improve model accuracy and predictive capabilities.
- **Scalable AI Infrastructure:** Leveraging cloud computing and big data frameworks will enable the integration of high-frequency sensor data at scale, facilitating automated decision-making for next-generation drilling operations.

Overall, the findings demonstrate that AI-driven methodologies, combined with strategic feature selection and ensemble ML modeling, provide a robust framework for predictive well control and integrity management. Implementing these recommendations will extend the applicability of this approach across the industry, supporting safer, more efficient, and data-driven drilling practices.

## Acknowledgement

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